← Back **Time to Practice: Self-Supervised Learning**
Graded Assignment • 15 min

Due Oct 28, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. What is the primary goal of self-supervised learning in the context of machine learning? 1 / 1 point

- To enhance the accuracy of supervised models
- To increase computational efficiency
- To simplify model architectures
- To reduce the dependency on labeled data

Nice work

The primary goal of self-supervised learning is to reduce the dependency on expensive labeled data by leveraging unlabeled data to generate pseudo-labels for training.

2. What is a common method used in self-supervised learning to create a prediction task? 1 / 1 point

- Removing part of the input
- Using external datasets
- Applying supervised systems
- Enhancing data resolution

Nice work

A typical approach in self-supervised learning is to occlude or remove part of the input and use the visible parts to predict the occluded parts.

3. A machine learning model is trained to predict the next frame in a video sequence. What type of learning is being utilized? 1 / 1 point

- Supervised learning
- Unsupervised learning
- Clustering
- Self-supervised learning

Nice work

This scenario involves predicting a future part of the input (the next frame of the video) based on the past frames, which aligns with the concept of self-supervised learning where the model predicts any part of the input from any other part.

4. What is transfer learning primarily used for in deep learning? 1 / 1 point

- Enhance data storage efficiency
- Optimize hardware resources
- Increase computational speed
- Reuse a learned feature representation for a related task

Nice work

Transfer learning allows the reuse of a pre-trained model on one task to be used on a new, but related task, leveraging the learned features.

5. Why is self-supervised learning particularly important for deep learning methods? 1 / 1 point

- Increases computational speed
- Simplifies model architectures
- Enhances model accuracy
- Reduces need for labeled data

Nice work

Deep learning methods require large amounts of training data because of their large number of parameters. Self-supervised learning helps in reducing the reliance on large amounts of expensive labeled data by utilizing unlabeled data effectively to produce effective training.